

Worksheet: Hamiltonian Circuits & Sorted Edges / Cheapest Link

Group Names: Solutions

Sorted Edges / Cheapest Link: Sort the edges from cheapest to most expensive. Add the next cheapest edge to your circuit *unless* (1) it closes the circuit too soon, or (2) it creates a degree 3 vertex. Break ties by choosing the alphabetically smallest edge.

1. In each picture, the thick edges have already been chosen by the Sorted Edges algorithm, and the dashed edge is the next cheapest edge. Should the dashed edge be added? If not, which rule does it break?

(a)

No! Creates a small circuit.

(b)

No! Creates a degree-3 vertex.

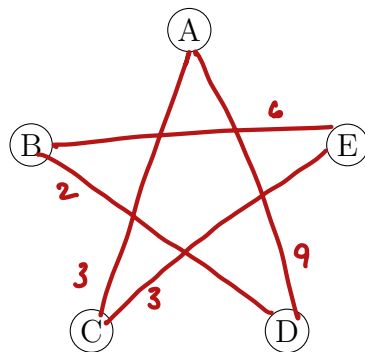
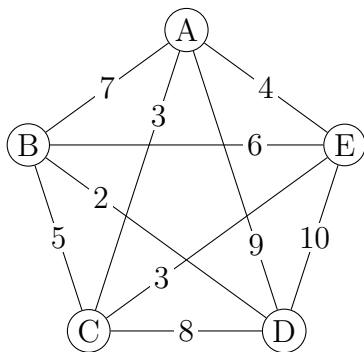
(c)

Yes - no problem

(d)

Yes! Finishes the circuit.

2. Apply the Sorted Edges algorithm to the graph below to find a Hamiltonian circuit. Draw in the edges, labeled with their weight, as you add them on the empty graph, and keep track of each step in the table. Break ties by choosing the alphabetically smaller edge.



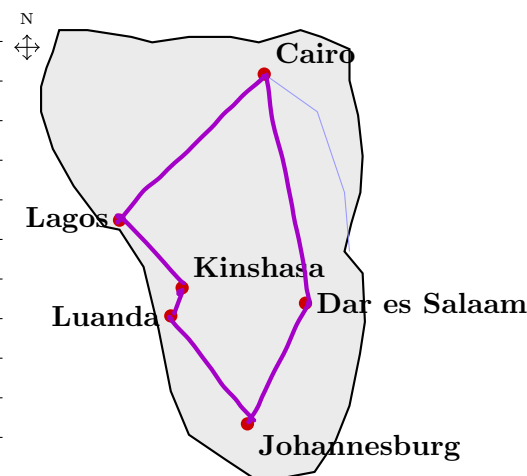
Sorted edges	weight	used? (or why not)
BD	2	✓
AC	3	✓
CE	3	✓
AE	4	✗ Creates circuit
BC	5	✗ creates deg 3
BE	6	✓
AB	7	✗ deg 3
CD	8	✗ deg 3
AD	9	✓ STOP
DE	10	

List the vertices of the Hamiltonian circuit, starting at vertex A.
ACEDBA

Total weight of the circuit:
23

3. On the previous worksheet, you used Nearest Neighbor to plan a tour of the six most populous cities in Africa. Here is the same data, with the edges already sorted by distance (in km).

Sorted edges	distance	used?/why not
Kinshasa – Luanda	554	✓
Lagos – Kinshasa	1,798	✓
Lagos – Luanda	2,027	X circuit
Dar es Salaam – Johannesburg	2,461	✓
Luanda – Johannesburg	2,484	✓
Kinshasa – Dar es Salaam	2,651	X circuit
Kinshasa – Johannesburg	2,781	X circuit
Luanda – Dar es Salaam	2,870	X circuit
Cairo – Lagos	3,913	✓
Cairo – Kinshasa	4,179	X circuit
Cairo – Dar es Salaam	4,184	✓ STOP
Lagos – Dar es Salaam	4,242	
Lagos – Johannesburg	4,508	
Cairo – Luanda	4,732	
Cairo – Johannesburg	6,264	



- (a) Use the Sorted Edges algorithm to find a Hamiltonian circuit. List the cities of your circuit in order, starting and ending at Cairo.

Cairo - Lagos - Kinshasa - Luanda - Johannesburg - Dar-es-Salaam

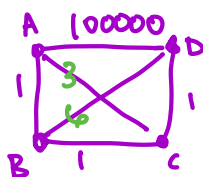
- (b) What is the total distance of your circuit? 15,394

- (c) Nearest Neighbor starting at Johannesburg gave a tour of 17,870 km. How does your Sorted Edges tour compare?

It's shorter!

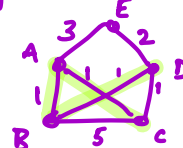
4. Does the Sorted Edges algorithm always find the Hamiltonian circuit with the smallest weight? Explain your reasoning, and give an example if you can.

No, it can force you to choose an expensive edge eventually, and it can also fail to be able to even find a HC on a non-complete graph



AB 1
BC 1
CD 1
AC 3
BD 6
AD 10000

} these 3 are used
so the only edge available to complete the circuit is the expensive one!



Choose:
AB and then can't complete, even though HC exists.